Example Worksheet: The Kitchen Sink

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July 17, 2026

A dummy worksheet demonstrating every construct the pipeline supports.

What you'll learn

Using this template

- Recognise every markup element the tex $\to$ MDX pipeline supports.
- Copy this file as the starting point for a new worksheet.

Structuring learning outcomes

- Group outcomes with `\subsection*{...}` plus an `itemize` when a flat list gets long.
- Drop the `\subsection*` headings for a short sheet – a single `itemize` is fine.

This dummy worksheet demonstrates every construct that survives the round trip from LaTeX to the curriculum website. Compile it with `pdflatex` (twice, then `bibtex`, then twice again for the bibliography), or just read the source side by side with the rendered page.

1 Prerequisites

Prerequisites, further reading, and anything else you want to say are just ordinary LaTeX — no special markup. For this sheet:

- comfort with probability on finite sets ($\mathbb{E}$, distributions);
- one variable of calculus (we differentiate $-t \log t$ once).

2 Prose, math, and cross-references

Ordinary prose converts verbatim, including *emphasis*, **bold**, `monospace`, and special characters like 50% or §3. Inline math is copied byte-for-byte: $e^{i\pi} + 1 = 0$, and KaTeX handles comparisons like $x_{<t}$ without trouble. Custom notation defined in the preamble just works: $D_{\mathrm{KL}}(p \,\|\, q) \geq 0$, $\arg\max_a \mathbb{E}[R \mid a]$, for p, q distributions on $\mathbb{R}^n$.

Display math works as

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$

and as numbered equations you can reference:

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1}$$

Multi-line derivations use `align`:

$$\begin{align} (a+b)^2 &= (a+b)(a+b) \tag{2} \\ &= a^2 + 2ab + b^2. \tag{3} \end{align}$$

Cross-references resolve to the exact text LaTeX prints: Equation (1) sits in Section 2; below we will meet Definition 3.1 and Theorem 3.2. Lists are fine too:

- first bullet;
- second bullet, with math 2^n.

3 Definitions, theorems, and callouts

Label anything you want to reference. Any placement LaTeX binds correctly is valid (top level of the environment); right after the `begin` is clearest.

Definition 3.1 (entropy)**.** The *entropy* of a distribution p on a finite set $\mathcal{X}$ is

$$H(p) = -\sum_{x \in \mathcal{X}} p(x) \log p(x).$$

Theorem 3.2 (Gibbs' inequality)**.** *For any distributions p, q on $\mathcal{X}$, $-\sum_x p(x) \log q(x) \geq H(p)$, with equality iff $p = q$.*

Proof. By definition, $D_{\mathrm{KL}}(p \,\|\, q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$ by Jensen's inequality applied to the concave function log; rearranging gives the claim. Equality in Jensen requires $p(x)/q(x)$ constant, i.e. $p = q$. (On the website this proof is collapsible, like a solution.) □

Lemma 3.3. *$H(p) \leq \log |\mathcal{X}|$, with equality iff p is uniform.*

Proof. Apply Theorem 3.2 with q uniform. □

Warning

Callouts flag pitfalls, tips, or side notes. Types: `note`, `tip`, `warning`. They may be labelled and referenced, like this one.

Remark

A `remark` is an aside; on the website it renders as a note-style callout. Referencing works: see Callout 3.1 above (and this remark is Remark 3.1).

4 Exercises and solutions

The optional argument of an exercise is its title, exactly as in amsthm. An optional `\important` mark (a ★ flagging the sheet's most important exercises) goes right after the label. Subparts are a plain `enumerate`; label an item to reference it. Every solution *names its exercise* — `[ex:label]` is mandatory — so you may place solutions right after each exercise or collect them at the end, as you prefer. Exercises don't require solutions; label any exercise a solution or reference points at.

Exercise 4.1 (Entropy warm-up). Let p be a distribution on a finite set $\mathcal{X}$.

(a) Show that $H(p) \geq 0$.

(b) For which p is $H(p) = 0$?

Solution to Exercise 4.1

For Exercise 4.1(a): each term $-p(x) \log p(x) \geq 0$ since $p(x) \in [0, 1]$. For part (b): $H(p) = 0$ iff every term vanishes iff p is a point mass. [*Hint:* when is $-t \log t = 0$ on $[0, 1]$?]

Exercise 4.2 (Gibbs, the hard direction). (★) Prove the equality case of Theorem 3.2 directly, without citing Jensen's equality condition. [*Note:* this one is solved at the end of the sheet instead of inline, to demonstrate both solution placements.]

Hints also have a block form: unnumbered, never labeled, rendered right where it is written — a bold lead-in on paper, a collapsible drop-down on the web.

Hint: Consider $f(t) = t - 1 - \log t$: it is nonnegative on $t > 0$ and vanishes only at $t = 1$.

5 Figures and citations

Figures use `figure` + `\includegraphics` (export TikZ to PDF the same way; keep assets in `fig/`).

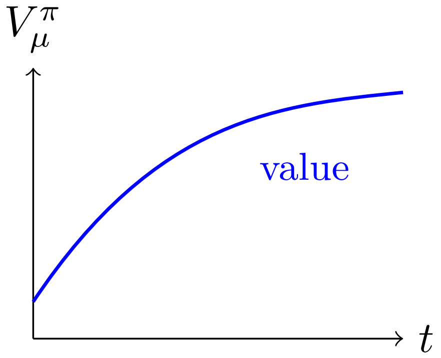

Figure 1: An example figure. On the web this becomes a `Figure` component.

See Figure 1. Citations come from the `.bib` file: the idea of entropy is due to [2]; for a textbook treatment see [1, ch. 4]. External links are plain hyperlinks, and footnotes are supported.[1]

Selected solutions

Solution to Exercise 4.2

Write $f(t) = t - 1 - \log t$ for $t > 0$ and note $f(t) \geq 0$ with equality iff $t = 1$. Then with $t_x = q(x)/p(x)$,

$$-\sum_x p(x) \log q(x) - H(p) = \sum_x p(x) f(t_x) + \sum_x \big(p(x) - q(x)\big) \geq 0,$$

and equality forces $t_x = 1$ wherever $p(x) > 0$, i.e. $p = q$.

References

[1] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.

[2] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.

[1] Footnotes render inline, in parentheses.